

1.1.3 Option C – Dryland Landscapes

1. How can dryland landscapes be viewed as systems?	
Key Ideas	Content
1.a. Dryland landscapes can be viewed as systems.	• A conceptual overview of: ◦ the components of dryland landscape systems, including inputs, processes and outputs ◦ the flows of energy and material through dryland systems ◦ aridity index, including UNEP.
1.b. Dryland landscapes are influenced by a range of physical factors.	• Potential influence on dryland systems of: ◦ climate, including precipitation totals and patterns ◦ geology, including lithology and structure ◦ latitude and altitude ◦ relief and aspect on microclimate ◦ the availability of sediment.
1.c. There are different types of dryland.	• The characteristics of different types of dryland landscapes: ◦ polar drylands ◦ mid- and low-latitude deserts ◦ semi-arid environments.

2. How are landforms of mid and low latitude deserts developed?	
Key Ideas	Content
2.a. Dryland landscapes develop due to a variety of interconnected climatic and geomorphic processes.	• The influence of flows of energy and materials on geomorphic processes, including weathering, mass movement, fluvial and aeolian erosion, transportation and deposition. • The formation of distinctive landforms, predominantly influenced by erosion, including wadis, canyons, pedestal rocks, ventifacts and desert pavements. • The formation of distinctive landforms, predominantly influenced by deposition, including barchans, linear dunes, star dunes, alluvial fans and bajadas.
2.b. Dryland landforms are inter-related and together make up characteristic landscapes.	• Case studies of one mid-latitude desert and one low-latitude desert to illustrate: ◦ the physical factors which influence the formation of landforms within the landscape system ◦ the inter-relationship of a range of landforms within the characteristic landscape system ◦ how and why the landscape system changes over time from millennia to seconds, such as the impact of flash floods on alluvial fans in seconds, seasonal and ephemeral streams on canyons and pediment development over the millennia.

3. How do dryland landforms evolve over time as climate changes?

Key Ideas	Content
3.a. Fluvial landforms can exist in dryland landscapes as a result of earlier pluvial periods.	• How dryland landforms have been influenced by previous pluvial conditions, including: ◦ climate changes that occurred during a previous time period and the resultant pluvial conditions ◦ the influence of pluvial conditions and geomorphic processes in shaping landforms, including inselbergs and pediments ◦ the modification of these landforms by processes associated with present and future climate changes.
3.b. Periglacial landforms can exist as a result of earlier colder periods.	• How dryland landscapes have been influenced by colder climatic conditions, including: ◦ climate changes that occurred during a previous time period and the resultant colder conditions ◦ the influence of colder climatic conditions and geomorphic processes in shaping landforms, including frost shattered debris, nivation hollows and solifluction deposits ◦ the modification of these landforms by processes associated with present and future climate changes.

4. How does human activity cause change within dryland landscape systems?

Key Ideas	Content
4.a. Water supply issues can cause change within dryland landscape systems.	• Case study of one dryland landscape that is being used by people, to illustrate: ◦ the water supply issue taking place and the reasons for it taking place, such as water shortage due to drought ◦ its impacts on processes and flows of material, processes and/or energy through the dryland landscape system, such as high rates of sediment trapping behind dams or modifying rivers to distribute and store water ◦ the effect of these impacts in changing dryland landforms, such as decreased growth of wadis ◦ the consequence of these changes on the landscape, such as reducing depositional landforms such as alluvial fans or slowing pediment development.

4. How does human activity cause change within dryland landscape systems?

Key Ideas	Content
4.b. Economic activity can cause change within dryland landscape systems.	• Case study of one dryland landscape that is being used by people, to illustrate: ◦ the economic activity taking place and the reasons for it taking place, such as tourism ◦ its impacts on processes and flows of material and/or energy through the dryland landscape system, such as vegetation and cryptobiotic crust damage by dune buggy use ◦ the effect of these impacts in changing dryland landforms, such as higher erosion rates on dunes ◦ the consequence of these changes on the landscape, such as increased loess accumulation in marginal areas.

5. Topic-specific skills:

- observation skills
- measurement and geo-spatial mapping skills
- data manipulation and statistical skills applied to field measurements
- sediment budget calculations
- mass balance calculations.